

M. MARUF

Applied Scientist | Foundation Models | Speech AI | Multimodal Generative AI

Seattle, WA

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Professional Summary

Applied Scientist at Amazon AGI specializing in multimodal foundation models, speech AI, large language model (LLM)-based systems, and generative AI. My work spans large-scale multimodal pretraining and post-training, including preference optimization and reinforcement learning, automatic speech recognition (ASR), text-to-speech synthesis (TTS), conversational AI, multilingual AI systems, and scalable evaluation infrastructure for foundation models.

Received a Ph.D. in Computer Science from Virginia Tech, where my research focused on multimodal foundational models, vision-language models, knowledge-guided machine learning, graph representation learning, generative modeling, and AI for scientific discovery. My research contributions have been published in leading venues including NeurIPS, ICLR, CVPR, ECCV, AACL, KDD, ICASSP, and SDM.

Industrial Experience

Amazon AGI

Applied Scientist II, AGI Foundations – Speech & Audio

Seattle, WA

Dec 2024 – Present

Worked on Nova Sonic speech-to-speech foundation models with a focus on pretraining, supervised fine-tuning, preference optimization, multilingual ASR and TTS evaluation, production training, and model delivery.

- **Pretraining:** Designed and improved large-scale pretraining recipes for multilingual speech foundation models, including data-mix strategy, rampdown schedules, ablation experiments, scaling ladder, and model architecture/search analysis for large models.
- **Post-training:** Generated persona-conditioned, multi-turn user-agent conversations covering challenging alphanumeric scenarios, synthesized them into speech, curated the resulting dataset, and incorporated it into the supervised fine-tuning mixture with a controlled sampling weight, improving performance on targeted alphanumeric evaluations. Ran Direct Preference Optimization using speaker-similarity and word-error-rate-based preference signals, improving speaker coherence and consistency in generated speech. Also designed verifiable reward functions for reinforcement learning and experimented with RL Gym-style workflows to improve the tool-calling capabilities of open-source models.
- **Scalable data generation and filtering:** Built data-quality and data-readiness pipelines for pretraining, including transcription filtering, spoken/written-form analysis, multilingual dataset preparation, synthetic data generation, and data normalization.
- **Evaluation and internal benchmarks:** Built scalable evaluation workflows and internal benchmarks for ASR/TTS quality, multilingual robustness, alphanumeric behavior, hallucination/error analysis, regression tracking, and production-readiness assessment.
- **Production impact:** Led and owned the release of the current Nova Sonic pretraining production checkpoint that shows approximately 14% ASR improvement and 67% TTS robust-WER improvement over the previous production baseline model across multilingual evaluation sets.

Amazon.com

Applied Scientist Intern

Bellevue, WA

May 2022 – Aug 2022

- Developed a real-time delivery image ranking algorithm to improve package delivery decision-making for Amazon drivers.
- Combined object-detection models with statistical ranking approaches, achieving approximately 70% improvement over prior baselines.

- Improved graph neural network models for GPS-based satellite position estimation in collaboration with the GNSS Location team.
- Investigated robust machine learning methods for localization under noisy satellite positioning signals.

Selected Publications

** indicates equal contribution. Full list of publication: [Google Scholar](#)*

Conference Proceedings and Preprints

1. **M. Maruf**, A. Daw, K. Mehrab, A. Karpatne, et al., “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images,” *NeurIPS*, 2024.
2. J. Fan, J. Li, **M. Maruf**, et al., “Can We Read the Mind of Audio LLMs? A Verbalizable, Multilingual Middle-Layer Workspace.” *under submission*, 2026
3. H. Manogaran, **M. Maruf**, A. Daw, A. Karpatne, et al., “What Do You See in Common? Learning Hierarchical Prototypes over Tree-of-Life to Discover Evolutionary Traits,” *ICLR*, 2025.
4. K. Mehrab*, **M. Maruf***, A. Daw*, A. Karpatne, et al., “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images,” *CVPR*, 2025.
5. Xinlu He, Swayambhu Nath Ray, **M. Maruf**, et al., “Continuous-token Diffusion for Speaker-Referenced TTS in Multimodal LLMs,” *ICASSP*, 2026.
6. M. Khurana, A. Daw, **M. Maruf**, A. Karpatne, et al., “Hierarchical Conditioning of Diffusion Models Using Tree-of-Life for Studying Species Evolution,” *ECCV*, 2024.
7. J. Bu*, A. Daw*, **M. Maruf***, and A. Karpatne, “Learning Compact Representations of Neural Networks using Discriminative Masking (DAM),” *NeurIPS*, 2021.
8. **M. Maruf***, A. Daw*, and A. Karpatne, “PID-GAN: A GAN Framework based on a Physics-informed Discriminator for Uncertainty Quantification with Physics,” *KDD*, 2021.
9. **M. Maruf** and A. Karpatne, “Maximizing Cohesion and Separation in Graph Representation Learning: A Distance-aware Negative Sampling Approach,” *SDM*, 2021.
10. **M. Maruf**, A. Daw, J. Bu, and A. Karpatne, “Beyond Discriminative Regions: Saliency Maps as Alternatives to CAMs for Weakly Supervised Semantic Segmentation,” *arXiv preprint arXiv:2308.11052*.

Journal Articles

1. M. Balk, J. Bradley, **M. Maruf**, A. Karpatne, et al., “A FAIR and Modular Image-based Workflow for Knowledge Discovery in the Emerging Field of Imageomics,” *Methods in Ecology and Evolution*, 2024.
2. Y. Bakis, Y. Altintas, **M. Maruf**, A. Karpatne, et al., “Extracting Landmark and Trait Information from Segmented Digital Specimen Images Generated by Artificial Neural Networks,” *Biodiversity Information Science and Standards*, 2022.
3. **M. Maruf** and S. Shatabda, “iRSpot-SF: Prediction of Recombination Hotspots by Incorporating Sequence-based Features into Pseudo Components,” *Genomics*, 2019.

Workshop Proceedings

1. **M. Maruf**, A. Daw, and A. Karpatne, “Are Pre-trained Vision Language Models (VLMs) Decent Zero-shot Predictors in Scientific Contexts?” *Imageomics Workshop, AAAI*, 2024.
2. K. Mehrab, A. Daw, **M. Maruf**, and A. Karpatne, “Phylo-GNN: Phylogeny-guided Graph Neural Network Approach for Fine-Grained Image Trait Identification,” *Imageomics Workshop, AAAI*, 2024.

3. A. Daw*, **M. Maruf***, and A. Karpatne, “PID-GAN: A GAN Framework based on a Physics-informed Discriminator for Uncertainty Quantification with Physics,” *SGAI-AAAI*, 2021.
4. **M. Maruf**, M. Elhamod, P. Mandke, and A. Karpatne, “Biology-guided Neural Network for Fish Trait Discovery,” *Integrative and Comparative Biology*, 2021.
5. **M. Maruf** and A. Karpatne, “Informing Neural Networks for Drug Effect Prediction Using Biological Knowledge of Protein-Protein Interactions,” *PIML*, 2020.

Professional Service and Scholarly Reviewing

- Reviewer for major AI conferences and workshops including NeurIPS, ICLR, ICML, CVPR, ECCV, NeurIPS Datasets and Benchmarks Track, KDD, PKDD, ML4PS-NeurIPS, and SGA-IAAI.
- Program Committee Member: AAAI 2026, NeurIPS 2024, SGA-IAAI (2021, 2022).
- Session Chair: Spatiotemporal Data Session, SDM 2021.

Education

Virginia Tech <i>Ph.D. in Computer Science, Advisor: Dr. Anuj Karpatne</i>	Blacksburg, VA <i>Dec 2024</i>
– Thesis: Learning without Expert Labels for Multimodal Data	
Virginia Tech <i>Master of Science in Computer Science</i>	Blacksburg, VA <i>May 2023</i>
Bangladesh University of Engineering and Technology (BUET) <i>Bachelor of Science in Computer Science and Engineering</i>	Dhaka, Bangladesh <i>Mar 2016</i>

Honors and Awards

- Best Ph.D. Researcher Award, Department of Computer Science, Virginia Tech, 2024.
- Kafura Graduate Fellowship, Virginia Tech, 2023.
- SDM Doctoral Student Travel Award, 2020.
- Dean’s List Award, Bangladesh University of Engineering and Technology (BUET).

Technical Expertise

Foundation Models	Large Language Models, Speech Foundation Models, Vision-Language Models, Multimodal Foundation Models, Diffusion Models, Generative AI
Pretraining and Post-training	Pretraining Recipe Design, Data-Mixture Optimization, Scaling and Ablation Studies, Supervised Fine-Tuning, Direct Preference Optimization, Reinforcement Learning, RL Gym, Synthetic Data Generation, Reward and Preference Signal Design
Speech and Multimodal AI	Speech-to-Speech Models, Automatic Speech Recognition, Text-to-Speech Synthesis, Speaker-Referenced Generation, Speaker Similarity, Multilingual Modeling, Conditional Generative Modeling
Training and Evaluation	Distributed Training, Data Generation and Filtering, Model Evaluation, Internal Benchmark Development, ASR/TTS Error Analysis, Production Model Selection, Scalable and Trustworthy AI Systems
Additional Research Areas	Knowledge-Guided Machine Learning, Graph Neural Networks, AI for Science, Scientific Machine Learning, AI for Accessibility
Frameworks	PyTorch, TensorFlow, JAX, Hugging Face, NeMo, DeepSpeed, LoRA